overview

Date	27 JUNE 2026
Team ID	SWTID-2026-6540
Project Name	**Opti-Crop: Smart Agricultural Production Optimization Engine**
Maximum Marks	

Project Overview:

OptiCrop: Smart Agricultural Production Optimization Engine is an AIpowered crop recommendation system designed to help farmers select the most suitable crop based on soil nutrients and environmental conditions. The application uses a machine learning model to analyze key agricultural parameters, including Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall, and recommends the crop that is most likely to achieve better productivity.
The system is developed using **Python** and the **Flask** framework for the backend, while **HTML**, **CSS**, and **JavaScript** are used to build a responsive and user-friendly web interface. The machine learning model is trained using agricultural data with libraries such as **Scikit-learn**, **Pandas**, and **NumPy**, and is stored as a model.pkl file for real-time predictions.

Users simply enter the required soil and environmental values through the web application. The Flask backend validates the input, passes it to the trained machine learning model, and displays the recommended crop instantly. This enables farmers to make informed, data-driven decisions instead of relying solely on traditional farming practices.

OptiCrop aims to improve agricultural productivity, optimize the use of natural resources, reduce the risk of crop failure, and promote sustainable farming practices. The project demonstrates how machine learning and web technologies can be combined to provide an intelligent, practical, and accessible solution for modern agriculture.